EMPLOYEE PAYROLL AND DEDUCTION ANALYSIS

Python Programming Case Study

1. Problem Statement

Organizations calculate employee salaries using basic pay, allowances, overtime, and incentives. Progressive deductions must then be applied to determine the net salary. Manual calculations can cause errors, so a program is needed to calculate gross salary, apply deductions, validate inputs, and generate a clear salary statement.

2. Objective

To develop a Python program that calculates gross salary from employee pay components, applies progressive deductions, validates input values, and displays a formatted salary statement.

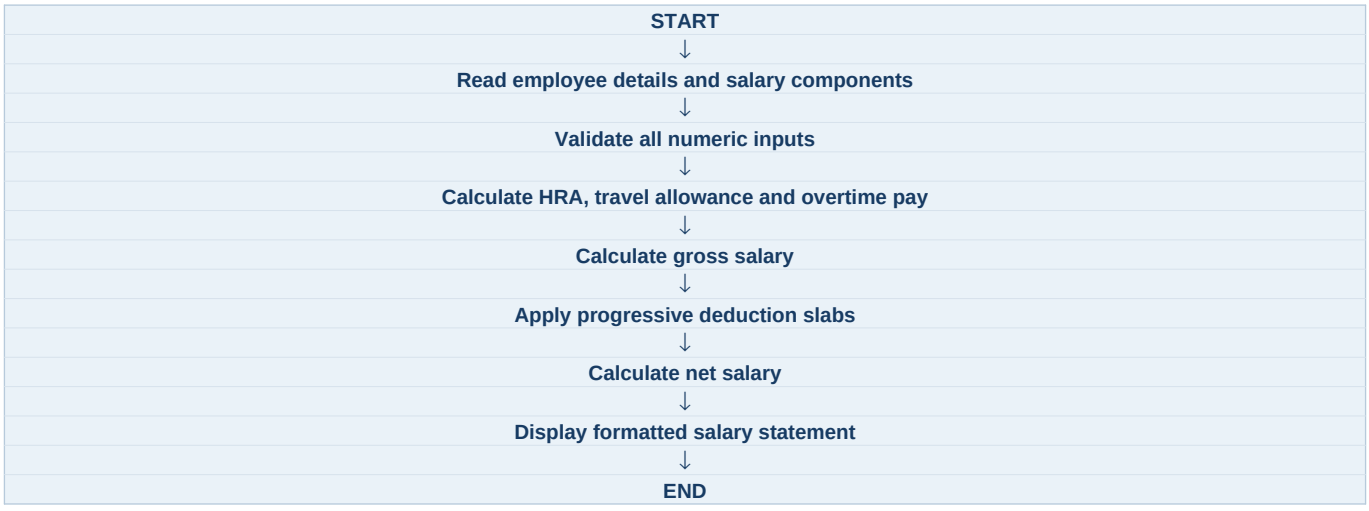
3. Assumptions

- Basic pay, allowances, overtime hours, overtime rate, and incentives must be non-negative.
- House Rent Allowance (HRA) is 20% of basic pay and travel allowance is 10% of basic pay.
- Overtime pay = overtime hours × overtime rate.
- Gross salary = basic pay + HRA + travel allowance + overtime pay + incentives.
- Progressive deductions are calculated on gross salary: 0% on the first ■20,000; 5% on the portion from ■20,001 to ■40,000; 10% on the portion from ■40,001 to ■60,000; and 15% on the portion above ■60,000.
- Net salary = gross salary – total deductions.
- The deduction slabs are sample rules for this programming exercise, not official tax or payroll advice.

4. Algorithm

1. Start the program and read employee name and employee ID.
2. Read basic pay, overtime hours, overtime rate, and incentives.
3. Validate that all entered numeric values are non-negative.
4. Calculate HRA and travel allowance as percentages of basic pay.
5. Calculate overtime pay by multiplying overtime hours by overtime rate.
6. Add basic pay, allowances, overtime pay, and incentives to calculate gross salary.
7. Calculate deductions progressively using the given salary slabs.
8. Subtract total deductions from gross salary to calculate net salary.
9. Display a formatted salary statement and stop.

5. Flowchart



6. Pseudocode

```
BEGIN
    INPUT employee_name, employee_id
    INPUT basic_pay, overtime_hours, overtime_rate, incentives

    IF any numeric input < 0
        DISPLAY "Invalid input"
        STOP
    END IF

    hra = basic_pay * 0.20
    travel = basic_pay * 0.10
    overtime_pay = overtime_hours * overtime_rate

    gross = basic_pay + hra + travel
            + overtime_pay + incentives
    deduction = 0
```

```
    IF gross > 20000
        deduction = deduction + (MIN(gross, 40000) - 20000) * 0.05
    END IF
    IF gross > 40000
        deduction = deduction + (MIN(gross, 60000) - 40000) * 0.10
    END IF
    IF gross > 60000
        deduction = deduction + (gross - 60000) * 0.15
    END IF

    net_salary = gross - deduction

    DISPLAY formatted salary statement
END
```

7. Python Code

```
name = input("Enter employee name: ")
emp_id = input("Enter employee ID: ")

basic = float(input("Enter basic pay: "))
ot_hours = float(input("Enter overtime hours: "))
ot_rate = float(input("Enter overtime rate per hour: "))
incentives = float(input("Enter incentives: "))

if basic < 0 or ot_hours < 0 or ot_rate < 0 or incentives < 0:
    print("Invalid input. Values cannot be negative.")
else:
    hra = basic * 0.20
    travel = basic * 0.10
    overtime_pay = ot_hours * ot_rate

    gross = basic + hra + travel + overtime_pay + incentives

    # Progressive deduction calculation
    deduction = 0
    if gross > 20000:
        deduction += (min(gross, 40000) - 20000) * 0.05
    if gross > 40000:
        deduction += (min(gross, 60000) - 40000) * 0.10
    if gross > 60000:
        deduction += (gross - 60000) * 0.15

    net = gross - deduction

    print("\\n========== SALARY STATEMENT ==========")
    print("Employee Name :", name)
    print("Employee ID   :", emp_id)
    print("--------------------------------------")
    print("Basic Pay     : INR", round(basic, 2))
    print("HRA           : INR", round(hra, 2))
    print("Travel Allow. : INR", round(travel, 2))
    print("Overtime Pay  : INR", round(overtime_pay, 2))
    print("Incentives    : INR", round(incentives, 2))
    print("--------------------------------------")
    print("Gross Salary  : INR", round(gross, 2))
    print("Deductions    : INR", round(deduction, 2))
    print("Net Salary    : INR", round(net, 2))
    print("======================================")
```

8. Expected Output (Illustration)

```
========== SALARY STATEMENT ==========
Employee Name : Ravi
Employee ID   : E101
--------------------------------------
Basic Pay     : INR 30000.0
HRA           : INR 6000.0
Travel Allow. : INR 3000.0
Overtime Pay  : INR 2000.0
Incentives    : INR 3000.0
--------------------------------------
Gross Salary  : INR 44000.0
Deductions    : INR 2000.0
Net Salary    : INR 42000.0
======================================
```